The energy that becomes thermal

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Account for mechanical energy changes through heating and deformation.

SUCCESS CRITERIA - I can...

1. Track system boundaries in an energy balance.
2. Distinguish thermal energy from temperature.
3. Use energy differences to quantify a transfer.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “thermal energy” and “system” to explain your model or result. Sentence starters are optional.

thermal energy: internal energy associated with microscopic motion/interactions

system: the object or objects chosen for analysis

conserved: unchanged in total under the stated conditions

01 Bring back an earlier idea

A spring stores 3 J at a given compression. Within its Hooke’s-law range, find the energy at 5 times that compression.

Word help: compression = shortening or squeezing relative to an unstressed length

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $E_{mechanical,\ i} = E_{mechanical,\ f} + \Delta E_{thermal}$.
No other transfers in this model; include contacting materials in the system. Friction transfers mechanical energy to internal energy.

Worked reasoning: A 2 N friction force opposes a sliding box over 5 m. It does -10 J of work; in a suitable box-floor system this appears as a 10 J thermal-energy increase.

Your turn: Draw the force, displacement and energy transfer.
Name the system for your energy account.

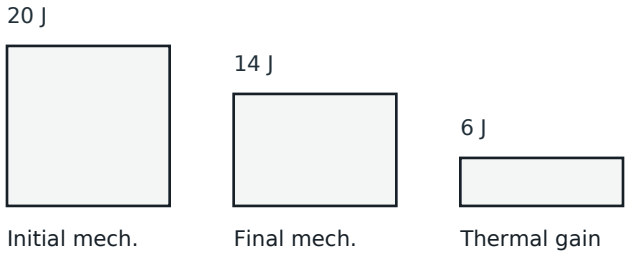
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Look and explain

A cart begins with 20 J mechanical energy and ends with 14 J. The closed accounting includes 6 J thermal increase. Does it balance?

Sentence starter: I notice ... . This supports ... because ... .



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

04 Build the solution

A 5 N friction force acts over 3 m.
Find work by friction and the modeled thermal energy increase for cart+surface.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

Does the same energy input always produce the same temperature rise in two different objects?
Name quantities you would need.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Choose a method

A cart’s work is measured over the first 4 m, but its speed change is measured over 6 m. Can these values directly test $W_{net}=\Delta K$?
Explain.

Word help: work = energy transferred by a force along a displacement

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

07 Exit check

An energy account has 12 J initially, 7 J motion energy and 3 J thermal energy finally. What is unaccounted for?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A box starts with 20 J of kinetic energy and stops under 4 N friction.
Find stopping distance and thermal-energy increase.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3